

Source: EPEX SDAC day-ahead auctions via Energy-Charts (Fraunhofer ISE, CC BY 4.0). Backtest by author.

TRADABLE SHARPE

**1.43**

after costs & lag

BOOTSTRAP 90% CI

**1.1–1.8**

block bootstrap

P&L / TRADED DAY

**+6.8**

€/MWh

PAPER SHARPE

**2.42**

untradable ceiling

MAX DRAWDOWN

**-97**

€/MWh

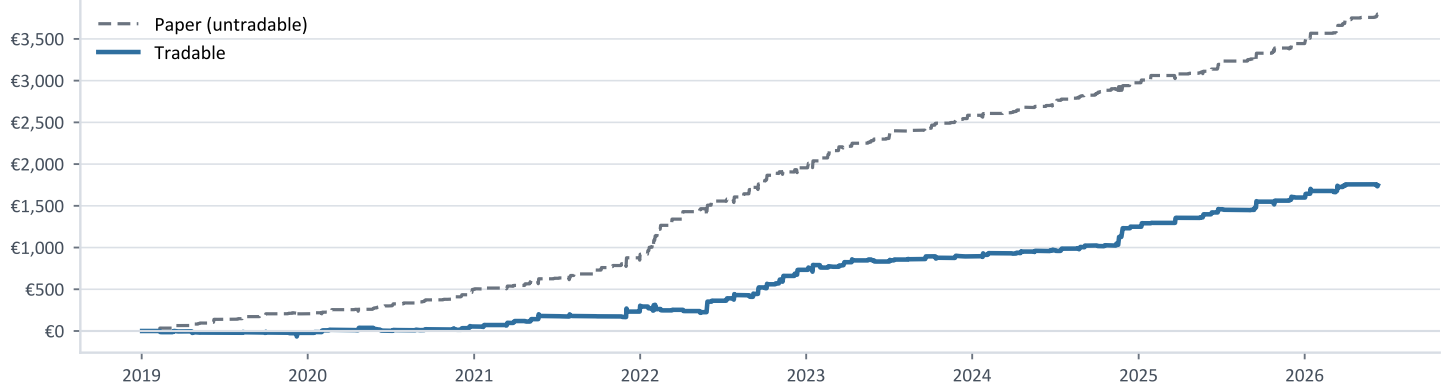
## STRATEGY & EDGE

CZ and DE-LU are joined by flow-based market coupling, so their day-ahead prices are usually identical and decouple only when the interconnector binds — typically afternoon hours under high renewables. The strategy fades those dislocations: when the daily average spread is more than 1.5 standard deviations from its trailing 30-day mean, it bets on reversion to coupling.

This is a statistical edge, not a structural flow, so it is presented with deliberate scepticism. Two variants are shown: a PAPER variant that (incorrectly) transacts at the prior day's printed price, and a TRADABLE variant that enters at the next biddable auction and unwinds a day later. The gap between them is the honesty check.

## Cumulative P&L: paper ceiling vs tradable implementation

EUR per MWh of spread position



## PAPER VS TRADABLE — WHY THE GAP

- Signal: z-score of the latest printed daily spread vs a 30-day rolling mean/ $\sigma$ ; trade when  $|z| > 1.5$ .
- Paper books the spread change from the prior day's print — a price that had already cleared. No instrument fills there.
- Tradable enters at the first auction still open after the signal and unwinds one day later, at 2 €/MWh round trip.
- The one-day execution lag plus costs roughly halves performance: Sharpe 2.42  $\rightarrow$  1.43.
- ~100% of P&L is the short side — fading rich-CZ congestion blowouts back to parity.
- Block bootstrap (2,000 draws, 10-day blocks) keeps the Sharpe positive across the 90% CI;  $P(\text{mean} \leq 0) = 0.000$ .

## HONEST HEALTH WARNING

Even the tradable Sharpe of 1.43 is an UPPER bound. Paired zonal day-ahead positions require a Balance Responsible Party in both zones, or thin PXE/EEX futures whose basis adds noise; and the quoted costs exclude the liquidity hole that opens exactly when the spread dislocates — when you most want to trade. Treat this as a research lead, not a production strategy.

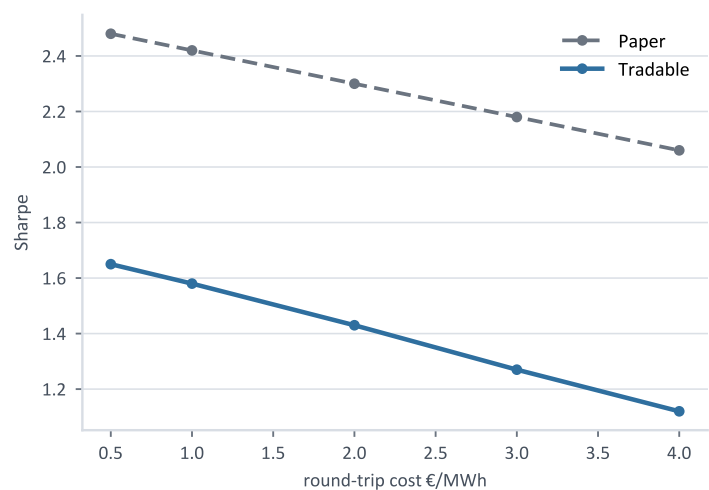
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## PAPER VS TRADABLE

| Metric             | Paper       | Tradable    |
|--------------------|-------------|-------------|
| Days in position   | 283         | 283         |
| Total P&L €/MWh    | 3,797       | 1,746       |
| P&L / traded day   | 13.75       | 6.83        |
| <b>Ann. Sharpe</b> | <b>2.42</b> | <b>1.43</b> |
| t-statistic        | 7.96        | 4.69        |
| Max drawdown       | -93         | -97         |
| Short-side P&L     | 3,334       | 1,744       |

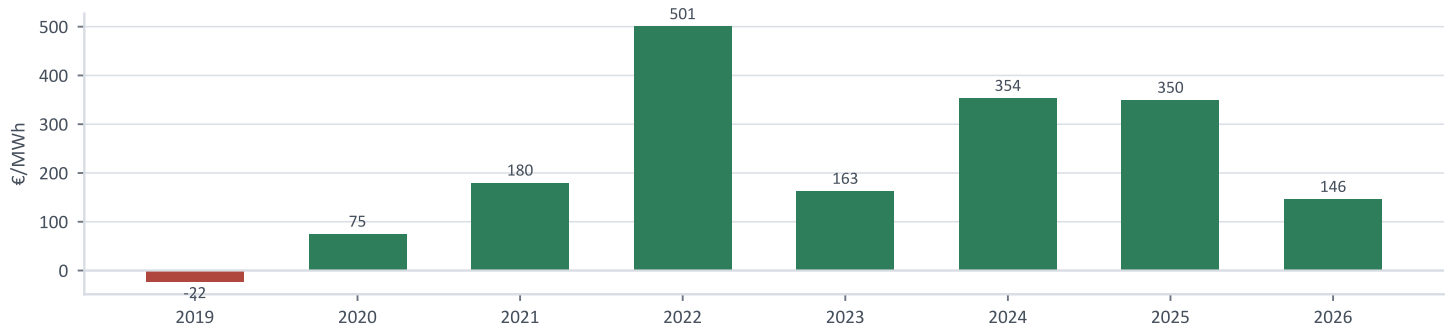
## Sharpe vs round-trip cost

annualised



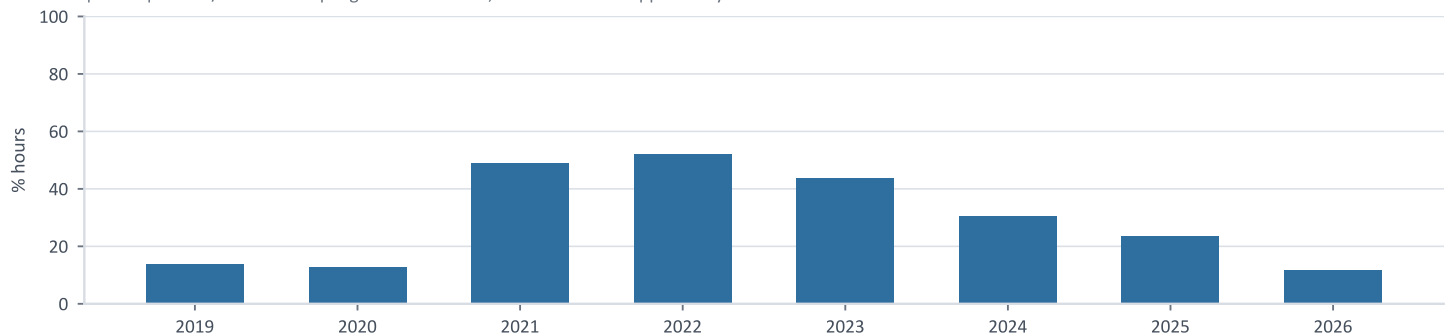
## Tradable P&L by year

EUR/MWh; 2019 was a losing year — no smooth equity line



## Spread behaviour: % of hours coupled

|CZ–DE| < 0.5 €/MWh — coupling is the base case, dislocations the opportunity



## KEY RISKS

Regime, not law: CZ has run at a structural premium to DE since 2022, so part of the 'reversion' is really riding a persistent level that could shift with new interconnection or policy. · Crash correlation: the spread blows out precisely in stressed grids (the 2018 Nasdaq clearing default that burned €107m was this family of trade). · Liquidity vanishes at the moment of opportunity. · Short-vol payoff: size to the 99.9th percentile, never to  $\sigma$  — power spreads have effectively unbounded kurtosis.